



**Shore**

Student Number:

Set:

**Year 12**  
**Mathematics Extension 1**  
**Trial Examination**  
**2008**

**General Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

**Note:** Any time you have remaining should be spent revising your answers.

**Total marks – 84**

- Attempt Questions 1 – 7
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

**Total Marks – 84**  
**Attempt Questions 1–7**  
**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

**Question 1 (12 marks)** **Marks**

- (a) Find the size of the acute angle between the two lines  $y = 2x - 3$  and  $y = 4 - x$ . Leave your answer correct to the nearest degree. **2**

- (b) Find a primitive of  $\int \frac{5}{\sqrt{4 - x^2}} dx$ . **2**

- (c) A family of two parents and four children is being arranged in a line for a family photograph. **2**

Calculate the number of possible arrangements if the two parents are placed at opposite ends of the line.

- (d) Solve for  $x$ :  $\frac{x}{8 - x} \leq 1$ . **3**

- (e) Consider the polynomial given by  $P(x) = x^3 - 12x + 16$ .

- (i) Show that  $(x - 2)$  is a factor of  $P(x)$ . **1**

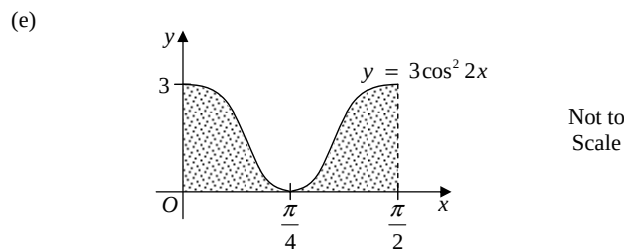
- (ii) Solve  $P(x) = 0$ . **2**

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

**Question 2 (12 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) Sketch the graph of  $y = 2\cos^{-1}x$ , clearly indicating the domain and range of the function. 2
- (b) How many different arrangements are there of the letters in the word NARRABRI? 2
- (c) Solve  $\sin 2x - \sin x = 0$  for  $0 \leq x \leq 2\pi$ . 2
- (d) Evaluate  $\int_0^1 x(x^2 + 1)^5 dx$  using the substitution  $u = x^2 + 1$ . 3



The diagram above shows the graph of  $y = 3\cos^2 2x$  for  $0 \leq x \leq \frac{\pi}{2}$ . Find the value of the shaded area bounded by the curve  $y = 3\cos^2 2x$ , the x-axis and the lines  $x = 0$  and  $x = \frac{\pi}{2}$ .

**Question 3 (12 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) A project team consisting of 2 engineers and 3 draftspersons is to be formed for a particular project. There are 6 engineers and 9 draftspersons available for selection. How many different project teams can be formed? 2

- (b) Use the expansion of  $\cos(A + B)$  to show that 2

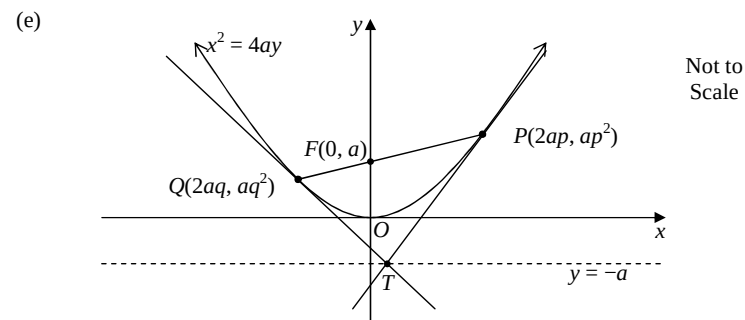
$$\cos 105^\circ = \frac{\sqrt{2} - \sqrt{6}}{4}.$$

- (c) A term of the expansion of  $(2 + mx)^{12}$  is  $24057x^5$ . Find the value of  $m$ . 2

- (d) Use the principle of mathematical induction to prove that 3

$$4 + 16 + 64 + \dots + 4^n = \frac{4^{n+1} - 4}{3}$$

for all integers  $n \geq 1$ .



The points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$  such that  $PQ$  is a focal chord with equation  $y = \left(\frac{p+q}{2}\right)x - apq$ . The focus,  $F$ , has coordinates  $(0, a)$  and  $O$  is the origin.

- (i) Show that  $pq = -1$ . 1
- (ii) The tangents to the parabola at  $P$  and  $Q$  have equations  $y = px - ap^2$  and  $y = qx - aq^2$  respectively. The tangents intersect at the point  $T$ . Show that  $T$  lies on the directrix  $y = -a$ . 2

**Question 4 (12 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) A particle is moving such that its displacement  $x$  metres from a fixed point  $O$  at any time  $t$  seconds is given by

$$x = 2 \cos 3t - \sqrt{12} \sin 3t.$$

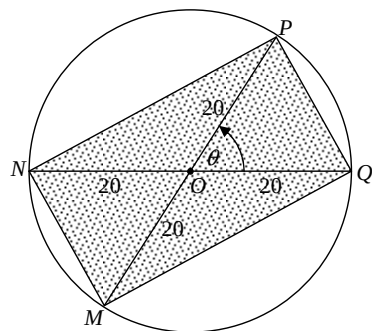
- (i) Prove that the motion is simple harmonic by showing that  $\ddot{x} = -9x$ .  
(ii) Find exact values for  $R$  and  $\alpha$  such that

$$2 \cos 3t - \sqrt{12} \sin 3t \equiv R \cos(3t + \alpha)$$

where  $R > 0$  and  $0 \leq \alpha \leq \frac{\pi}{2}$ .

- (iii) Hence, find the first time when the particle is at the centre of motion.

(b)



Not to Scale

A rectangle  $NPQM$  is inscribed in a circle of radius 20 centimetres with centre at  $O$  such that  $NQ$  and  $MP$  are diameters. The angle,  $\theta$  radians, between these two diameters is increasing at the rate of 0.1 radians per second.

- (i) Show that the area of the rectangle,  $A$  square centimetres, is given by

$$A = 800 \sin \theta.$$

- (ii) Calculate the rate at which the area  $A$  is changing when the angle between the diameters is  $\frac{\pi}{6}$  radians.

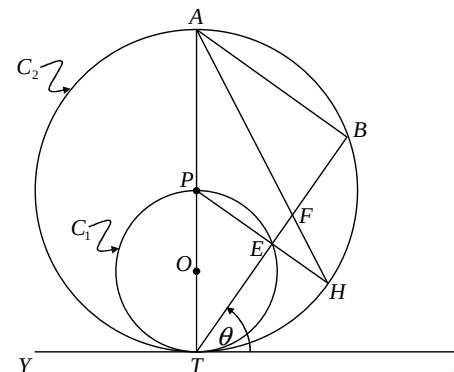
- (c) By applying the binomial theorem to  $(1+x)^{2n}$  and differentiating, show that

$$n \times 4^n = {}^{2n}C_1 + 2 {}^{2n}C_2 + \dots + k {}^{2n}C_k + \dots + 2n {}^{2n}C_{2n}.$$

**Question 5 (12 marks)** Use a SEPARATE writing booklet

**Marks**

(a)



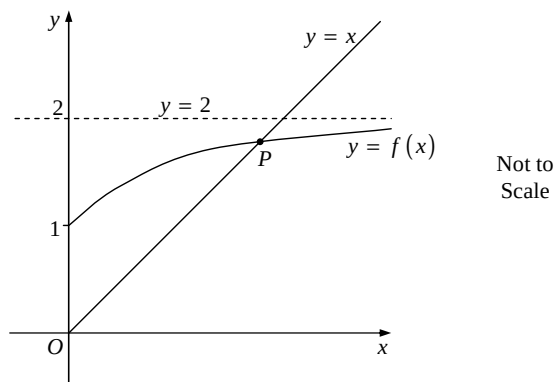
Not to Scale

In the diagram,  $O$  is the centre of the smaller circle  $C_1$ .  $P$  is a point on the circle  $C_1$  and the centre of the larger circle  $C_2$ .  $XY$  is a common tangent to both the circles at the point  $T$ . The diameter  $PT$  of circle  $C_1$  is produced to meet circle  $C_2$  at the point  $A$ . The points  $B$  and  $H$  lie on the circle  $C_2$ .  $BT$  meets circle  $C_1$  at the point  $E$  and  $AH$  at the point  $F$ . Let  $\angle BTX = \theta$ .

Show that  $AH$  bisects  $\angle TAB$ .

**Question 5 continues**

(b)



A function is defined as  $f(x) = 2 - e^{-x}$  for  $x \geq 0$ . The diagram shows the graph of  $y = f(x)$ , the lines  $y = x$  and  $y = 2$ . The curve  $y = f(x)$  and the line  $y = x$  intersect at the point  $P$ .

Copy or trace the diagram onto your writing booklet.

- (i) State the domain of the inverse function  $y = f^{-1}(x)$ . 1
- (ii) On your diagram, sketch the inverse function  $y = f^{-1}(x)$ . 2
- (iii) Show that the  $x$ -coordinate of the point  $P$  is a solution of the equation  $x + e^{-x} - 2 = 0$ . 1
- (iv) A first approximation to the solution of the equation  $x + e^{-x} - 2 = 0$  is  $x = 1.8$ . Use one application of Newton's method to find a better approximation of the  $x$ -coordinate of  $P$ . Give your answer correct to four decimal places. 2
- (v) Hence, approximate the area enclosed between the function  $y = f(x)$ , its inverse  $y = f^{-1}(x)$  and the coordinate axes. Leave your answer correct to 4 decimal places. 3

**End of Question 5**

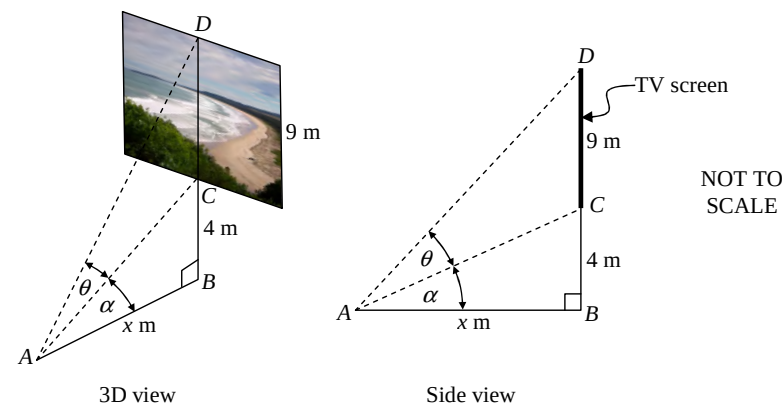
**Question 6 (12 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) A particle moves so that its acceleration at any time  $t$  seconds is given by  $\ddot{x} = -8e^{-4x}$ . Initially the particle is at the origin,  $O$ , with velocity 2 m/s.
  - (i) Show that the velocity of the particle at any time  $t$  is given by  $\dot{x} = 2e^{-2x}$ . 3
  - (ii) Hence, show that the displacement of the particle at any time  $t$  is given by  $x = \frac{1}{2} \ln(4t + 1)$ . 3

- (b) A rectangular TV screen is set up in a level field for spectators to watch a movie. The screen is mounted 4 metres off the ground and is 9 metres high and 16 metres wide.

Tom sits directly in front of the centre of the screen at the point  $A$  which is  $x$  metres from the point  $B$  directly below the screen as shown in the diagrams below. The angle  $DAC$  through which he views the bottom of the screen  $C$  and the top of the screen  $D$  is  $\theta$  radians as shown.



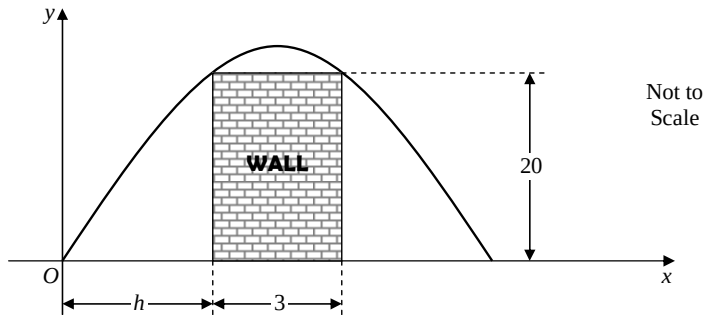
Let angle  $CAB$  be  $\alpha$  radians.

- (i) By considering the expansion of  $\tan(\theta + \alpha)$ , show that  $\theta = \tan^{-1}\left(\frac{9x}{52 + x^2}\right)$ . 3
- (ii) Hence, or otherwise, find the value of  $x$  (correct to 2 decimal places) for which Tom has the maximum viewing angle  $\theta$ . Justify your answer. 3

**Question 7 (12 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) A six-sided die is weighted so that the probability of rolling a “six” is twice that of any other number. Find the probability of rolling at least nine “sixes” in 10 rolls of this die. [You do not need to simplify your answer.] **3**
- (b) The wall of a fort on level ground is 3 metres thick and 20 metres high. A projectile is fired from a point  $O$  outside the fort,  $h$  metres from the base of the wall of the fort, towards the fort as shown in the diagram below.



It is assumed that the path of the projectile traces out a parabola of the form  $y = bx - ax^2$  where  $a$  and  $b$  are constants.

- (i) Show that  $b = \frac{20(2h + 3)}{h(h + 3)}$  and  $a = \frac{20}{h(h + 3)}$ . **3**  
 [i.e. Show that  $y = \frac{20(2h + 3)x}{h(h + 3)} - \frac{20x^2}{h(h + 3)}$ .]
- (ii) Let the angle of projection of the projectile be  $\theta$  degrees and the initial velocity be  $V$  m/s and the constant of gravity be  $g = 10$ . **1**  
 Hence the equations of motion are  $x = Vt \cos \theta$  and  $y = Vt \sin \theta - 5t^2$ .  
 Show that the equation of the path of flight of the projectile is given by
- $$y = x \tan \theta - \frac{5x^2}{V^2 \cos^2 \theta}.$$
- (iii) Hence, show that  $V^2 \cos^2 \theta = \frac{h(h + 3)}{4}$ . **1**
- (iv) If the projectile is fired at an angle of  $45^\circ$ , find the values of  $h$  and  $V$  correct to 2 decimal places. **4**

**End of Paper**

1a)  $m_1 = 2, m_2 = -1$

$$\tan \theta = \left| \frac{2 - (-1)}{1 + 2 \times (-1)} \right|$$

$$\tan \theta = \left| \frac{3}{-1} \right|$$

$$\theta = \tan^{-1} 3$$

$$\text{Angle} = 71^\circ 33' 54.18''$$

$$\approx 72^\circ \text{ (nearest degree)}$$

b)  $\int \frac{5}{\sqrt{4-x^2}} dx = 5 \int \frac{1}{\sqrt{4-x^2}} dx$   
 $= 5 \sin^{-1} \left( \frac{x}{2} \right) + C$

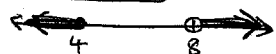
c)  $\boxed{2} \times \boxed{4} \times \boxed{3} \times \boxed{2} \times \boxed{1} \times \boxed{1}$   
 P C C C C P  
 Arrangements =  $2! \times 4!$   
 $= 48$

d)  $\frac{x}{8-x} \leq 1$   
critical points

1)  $\boxed{x=8}$

2)  $\frac{x}{8-x} = 1$   
 $x = 8-x$   
 $2x = 8$

$$\boxed{x=4}$$



$$\therefore x \leq 4 \text{ or } x > 8$$

① e) i)  $P(2) = 2^3 - 12 \times 2 + 16$   
 $= 0$

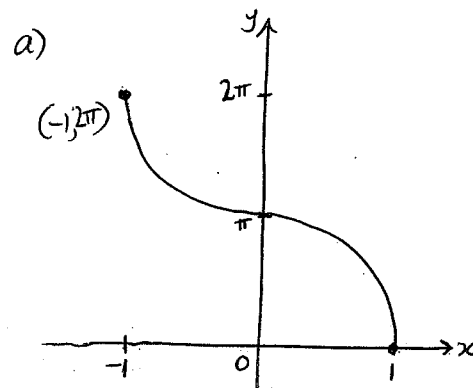
$\therefore (x-2)$  is a factor

ii)  $P(x) = (x-2)(x^2 + 2x - 8)$   
 $= (x-2)(x+4)(x-2)$   
 let  $P(x) = 0$

$$0 = (x-2)^2(x+4)$$

$$\therefore x = 2 \text{ or } x = -4$$

### Question 2



b) arrangements =  $\frac{8!}{3!2!}$   
 $= 3360$

c)  $\sin 2x - \sin x = 0$

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x (2 \cos x - 1) = 0$$

$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$

$$x = 0, \pi, 2\pi \text{ or } \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$\Rightarrow x = 0, \pi, 2\pi, \frac{\pi}{3} \text{ or } \frac{5\pi}{3}$$

② d)  $\int_0^1 x(x^2+1)^5 dx$

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

| limits |       |
|--------|-------|
| $x=0$  | $x=1$ |
| $u=1$  | $u=2$ |

$$\begin{aligned} \therefore \frac{1}{2} \int_0^1 2x(x^2+1)^5 dx &= \frac{1}{2} \int_1^2 u^5 du \\ &= \frac{1}{2} \left[ \frac{u^6}{6} \right]_1^2 \\ &= \frac{1}{2} \left( \frac{2^6}{6} - \frac{1^6}{6} \right) \\ &= \underline{\underline{5\frac{1}{4}}} \end{aligned}$$

e). Curve is symmetrical about  $x = \frac{\pi}{4}$ .

$$\begin{aligned} \therefore \text{Area} &= 2 \int_0^{\frac{\pi}{4}} 3 \cos^2 2x dx \\ &= 6 \int_0^{\frac{\pi}{4}} \cos^2 2x dx \end{aligned}$$

|                                                |
|------------------------------------------------|
| Consider                                       |
| $\cos 2A = 2\cos^2 A - 1$                      |
| $\therefore \cos^2 A = \frac{\cos 2A + 1}{2}$  |
| $\therefore \cos^2 2x = \frac{\cos 4x + 1}{2}$ |

2e) cont'd....

$$\text{Area} = 6 \int_0^{\frac{\pi}{4}} \frac{\cos 4x + 1}{2} dx$$

$$= 3 \int_0^{\frac{\pi}{4}} (\cos 4x + 1) dx$$

$$= 3 \left[ \frac{1}{4} \sin 4x + x \right]_0^{\frac{\pi}{4}}$$

$$= 3 \left( \left( \frac{1}{4} \sin \frac{\pi}{4} + \frac{\pi}{4} \right) - \left( \frac{1}{4} \sin 0 + 0 \right) \right)$$

$$= \underline{\underline{\frac{3\pi}{4} \text{ units}^2}}$$

### QUESTION 3

a). teams =  ${}^6C_2 \times {}^9C_3$

$$= \underline{\underline{1260}}$$

b).  $\cos(A+B) = \cos A \cos B - \sin A \sin B$

$$\begin{aligned} \therefore \cos 105^\circ &= \cos(60^\circ + 45^\circ) \\ &= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ &= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{1 - \sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \underline{\underline{\frac{\sqrt{2} - \sqrt{6}}{4}}} \end{aligned}$$

③ c)  $(2+mx)^{12}$

$$24057x^5 = {}^{12}C_5 2^7 (mx)^5$$

$$= 792 \times 2^7 m^5 x^5$$

$$\therefore 24057 = 101376 m^5$$

$$m^5 = \frac{243}{1024}$$

$$\therefore m = \frac{3}{4}$$

d) Prove true for  $n=1$

$$LHS = 4^1$$

$$= 4$$

$$RHS = \frac{4^{1+1} - 4}{3}$$

$$= \frac{12}{3}$$

$$= 4$$

$$= LHS$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\text{i.e. } 4 + 16 + 64 + \dots + 4^k = \frac{4^{k+1} - 4}{3}$$

Prove true for  $n=k+1$

$$\text{i.e. prove } 4 + 16 + 64 + \dots + 4^k + 4^{k+1} = \frac{4^{k+2} - 4}{3}$$

$$LHS = 4 + 16 + 64 + \dots + 4^k + 4^{k+1}$$

$$= \frac{4^{k+1} - 4}{3} + 4^{k+1} \quad \text{by assumption}$$

$$= \frac{4^{k+1} - 4 + 3 \cdot 4^{k+1}}{3}$$

$$= \frac{4 \cdot 4^{k+1} - 4}{3}$$

③ d) cont'd...

$$LHS = \frac{4^{k+2} - 4}{3}$$

$$= RHS$$

Hence, if it is true for  $n=k$ , then it is true for  $n=k+1$ .

Since it is true for  $n=1$ , then it is true for  $n=1+1=2$ , and also for  $n=2+1=3$ , and so on for all integers  $n \geq 1$ .

③ e) i) Since  $F(0, a)$  lies on focal chord, subst.

$(0, a)$  into eqn of chord.

$$a = \left(\frac{p+q}{2}\right) \times 0 - apq$$

$$\frac{a}{-a} = \frac{-apq}{-a}$$

$$pq = -1$$

ii) Solve tangents simultaneously

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$(p-q)x = a(p-q)(p+q)$$

$$x = a(p+q)$$

$$\text{sub into } y = px - ap^2$$

$$y = p(a(p+q)) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

but  $pq = -1$  (part (i))

$\therefore y = -a$  which means  $T$  lies on directrix.

# QUESTION 4

a) i)  $x = 2\cos 3t - \sqrt{2}\sin 3t$

$$\dot{x} = -6\sin 3t - 3\sqrt{2}\cos 3t$$

$$\ddot{x} = -18\cos 3t + 9\sqrt{2}\sin 3t$$

$$= -9(2\cos 3t - \sqrt{2}\sin 3t)$$

$$\therefore \ddot{x} = -9x \quad \text{as } x = 2\cos 3t - \sqrt{2}\sin 3t$$

ii) let  $2\cos 3t - \sqrt{2}\sin 3t \equiv R\cos(3t + \alpha)$

$$= R\cos 3t \cos \alpha - R\sin 3t \sin \alpha$$

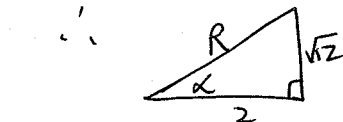
equating coefficients

$$R\cos \alpha = 2$$

$$\cos \alpha = \frac{2}{R}$$

$$R\sin \alpha = \sqrt{2}$$

$$\sin \alpha = \frac{\sqrt{2}}{R}$$



By pythag

$$R^2 = \sqrt{2}^2 + 2^2$$

$$\therefore R = 4$$

$$\tan \alpha = \frac{\sqrt{2}}{2}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore 2\cos 3t - \sqrt{2}\sin 3t \equiv 4\cos(3t + \frac{\pi}{3})$$

iii) At centre of motion when  $x = 0$

$$4\cos(3t + \frac{\pi}{3}) = 0$$

$$\cos(3t + \frac{\pi}{3}) = 0$$

$$\therefore 3t + \frac{\pi}{3} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$3t = \frac{\pi}{6}, \frac{7\pi}{6}, \dots$$

$$t = \frac{\pi}{18}, \dots$$

$\therefore$  First at centre of motion after  $\frac{\pi}{18}$  seconds.

4b) i). Area of  $\Delta POQ = \frac{1}{2} \times 20 \times 20 \sin \theta$   
 $= 200 \sin \theta$

$$\text{Area of } \Delta NOP = \frac{1}{2} \times 20 \times 20 \sin(180 - \theta)$$

$$= \frac{1}{2} \times 20 \times 20 \sin \theta \quad (\sin \theta = \sin(180 - \theta))$$

$$= 200 \sin \theta$$

$$\therefore \text{Area Rectangle} = 2 \times \text{Area } \Delta POQ + 2 \times \text{Area } \Delta NOP$$

$$= 2 \times 200 \sin \theta + 2 \times 200 \sin \theta$$

$$\therefore \underline{\underline{A = 800 \sin \theta}}$$

ii)  $\frac{d\theta}{dt} = 0.1$  (given)

$$\text{need } \frac{dA}{dt} = \frac{dA}{d\theta} \times \frac{d\theta}{dt}$$

$$\frac{dA}{dt} = 800 \cos \theta \times 0.1$$

$$= 80 \cos \theta$$

When  $\theta = \frac{\pi}{6}$

$$\frac{dA}{dt} = 80 \cos \frac{\pi}{6}$$

$$= \frac{80\sqrt{3}}{2}$$

$$\therefore \text{Area increasing at } \underline{\underline{40\sqrt{3} \text{ cm}^2/\text{second}}}$$

④ c)  $(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + {}^{2n}C_3 x^3 + \dots + {}^{2n}C_k x^k + \dots + {}^{2n}C_{2n} x^{2n}$

Differentiate both sides w.r.t.  $x$ .

$$2^n(1+x)^{2n-1} = 0 + {}^{2n}C_1 + 2 {}^{2n}C_2 x + 3 {}^{2n}C_3 x^2 + \dots + k {}^{2n}C_k x^{k-1} + \dots + {}^{2n}C_{2n} x^{2n-1}$$

Subst  $x=1$

$$2n(HI)^{2n-1} = 2^n C_1 + 2^{2n} C_2(I) + 3^{2n} C_3(I)^2 + \dots + k^{2n} C_k(I)^{k-1} + \dots + 2^{2n} C_{2n}(I)^{2n-1}$$

$$2^n x 2^{2^n-1} = 2^n C_1 + 2^{2^n} C_2 + 3^{2^n} C_3 + \dots + k^{2^n} C_k + \dots + 2^{2^n} C_{2^n}$$

$$n \times 2^1 \times 2^{2n-1} = 2^{2n} C_1 + 2^{2n} C_2 + 3^{2n} C_3 + \dots + 2n^{2n} C_{2n}$$

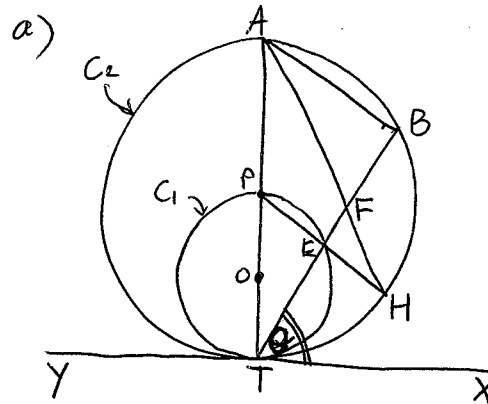
$$n \times 2^{2n} = \dots$$

$$n \times 4^n = {}^n C_1 + 2 {}^n C_2 + 3 {}^n C_3 + \dots + 2n {}^n C_{2n}$$

### QUESTION 5

see over...

### QUESTION 5



$\angle TAB = \angle BTX$   
 $= \theta$

In circle  $C_2$ ,  
 (angle between tangent and chord  
 drawn to point of contact is equal  
 to angle in alternate segment)

likewise in circle  $C_1$ ,

$$\angle TPH = \angle BTX = \theta.$$

Now,

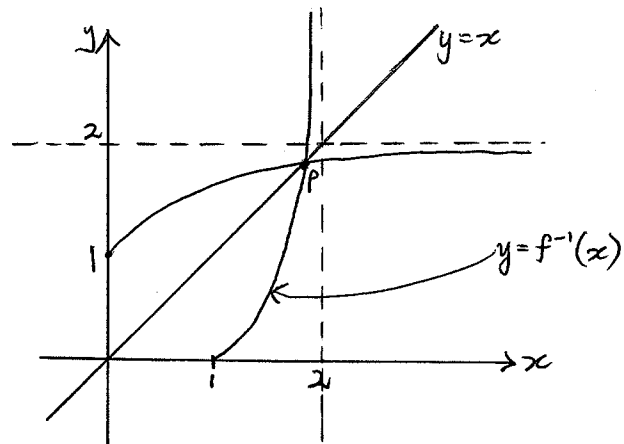
$\angle T P H = 2 \times \angle T A H$  (angle at centre of circle  $C_2$  is equal to twice angle at circumference standing on same arc)

$\therefore \angle T A H = \frac{\theta}{2}$

$$\therefore \angle TAH = \frac{1}{2} \times \angle TAB$$

$\therefore AH$  bisects  $\angle TAB$ .

⑤ b)



i) Domain of  $y=f^{-1}(x)$  is  $1 \leq x < 2$ .

ii) P lies on point of intersection of  $y=x$  and  $y=f(x)$

Solving simultaneously,

$$x = 2 - e^{-x}$$

$$\therefore x + e^{-x} - 2 = 0$$

iii) let  $g(x) = x + e^{-x} - 2$

$$\therefore g'(x) = 1 - e^{-x}$$

Better approx of x-coord of P

$$x_1 = 1.8 - \frac{1.8 + e^{-1.8} - 2}{1 - e^{-1.8}}$$

$$= 1.84157 \dots$$

$$\div 1.8416 \quad (4 \text{ dp})$$

⑤ b) cont'd...

$$v) \text{ Area} = 2 \times \int_0^{1.8416} (2 - e^{-x} - x) dx$$

$$= 2 \left[ 2x + e^{-x} - \frac{x^2}{2} \right]_0^{1.8416}$$

$$= 2 \left[ (2 \times 1.8416 + e^{-1.8416} - \frac{1.8416^2}{2}) - (0 + e^0 - 0) \right]$$

$$= 2 [ 2.146 \dots - 1 ]$$

$$= 2.29203 \dots$$

$$\div 2.2920 \text{ units}^2 \quad (4 \text{ dp})$$

## QUESTION 6

$$a) i) \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = \ddot{x}$$

$$\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -8e^{-4x}$$

$$\frac{1}{2} v^2 = \int -8e^{-4x} dx$$

$$\frac{1}{2} v^2 = 2e^{-4x} + C$$

$$\text{When } x=0, v=2$$

$$\frac{1}{2} \times 2^2 = 2e^0 + C$$

$$2 = 2 + C$$

$$0 = C$$

⑥ a) i) cont'd...

$$\therefore \frac{1}{2}v^2 = 2e^{-4x}$$

$$v^2 = 4e^{-4x}$$

$$v = \pm \sqrt{4e^{-4x}}$$

But when  $x=0$ ,  $v > 0$

$$\therefore v = \sqrt{4e^{-4x}}$$

$$= 2(e^{-4x})^{\frac{1}{2}}$$

$$\therefore v = 2e^{-2x}$$

⑥ a) ii)

$$\frac{dx}{dt} = 2e^{-2x}$$

$$\frac{dx}{dt} = \frac{2}{e^{2x}}$$

$$\frac{dt}{dx} = \frac{e^{2x}}{2}$$

$$t = \int \frac{e^{2x}}{2} dx$$

$$t = \frac{1}{2} \times \frac{1}{2} e^{2x} + c_1$$

When  $t=0$ ,  $x=0$

$$\therefore 0 = \frac{1}{4}e^0 + c_1$$

$$\therefore c_1 = -\frac{1}{4}$$

$$\text{Hence } t = \frac{1}{4}e^{2x} - \frac{1}{4}$$

$$4t = e^{2x} - 1$$

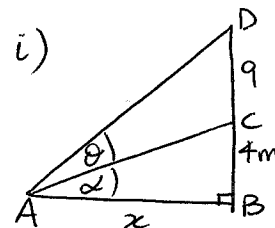
$$4t+1 = e^{2x}$$

⑥ a) ii) cont'd...

$$\log_e(4t+1) = 2x$$

$$\therefore x = \frac{1}{2} \ln(4t+1)$$

⑥ b) i)



$$\text{In } \triangle CAB, \tan \alpha = \frac{4}{x}$$

$$\text{In } \triangle BAD, \tan(\theta + \alpha) = \frac{13}{x}$$

$$\text{Consider } \tan(\theta + \alpha) = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha}$$

$$\therefore \frac{13}{x} = \frac{\tan \theta + \frac{4}{x}}{1 - \frac{4 \tan \theta}{x}} \times \frac{x}{x}$$

$$\frac{13}{x} = \frac{x \tan \theta + 4}{x - 4 \tan \theta}$$

$$13(x - 4 \tan \theta) = x(x \tan \theta + 4)$$

$$13x - 52 \tan \theta = x^2 \tan \theta + 4x$$

$$9x = x^2 \tan \theta + 52 \tan \theta$$

$$9x = \tan \theta (x^2 + 52)$$

$$\frac{9x}{x^2 + 52} = \tan \theta$$

$$\therefore \theta = \tan^{-1} \left( \frac{9x}{x^2 + 52} \right)$$

⑥ b) ii)

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{9x}{x^2+52}\right)^2} \times \frac{(x^2+52) \cdot 9 - 9x(2x)}{(x^2+52)^2}$$

Maximum value of  $\theta$  when  $\frac{d\theta}{dx} = 0$

$$0 = \frac{9(x^2+52) - 18x^2}{\left(1 + \left(\frac{9x}{x^2+52}\right)^2\right)(x^2+52)^2}$$

$$0 = 9x^2 + 468 - 18x^2$$

$$0 = 468 - 9x^2$$

$$= 9(52 - x^2)$$

$$\therefore x^2 = 52$$

$$x = \pm\sqrt{52}$$

But  $x$  is a length and must be positive.

$$\text{test } x = \sqrt{52}$$

|                      |          |             |        |
|----------------------|----------|-------------|--------|
|                      |          | 7.21        |        |
| $x$                  | 7        | $\sqrt{52}$ | 8      |
| $\frac{d\theta}{dx}$ | 0.101... | 0           | -0.004 |

$\therefore$  maximum viewing angle when  $x = \sqrt{52}$   
i.e.  $x \approx 7.21$   
(2dp)

## QUESTION 7

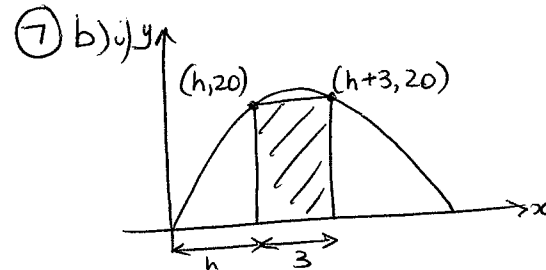
⑦ a)  $P(\text{one roll, one six}) = \frac{2}{7}$

$$P(\text{at least 9 sixes}) = {}^{10}C_{10}\left(\frac{2}{7}\right)^{10} + {}^{10}C_9\left(\frac{2}{7}\right)^9\left(\frac{5}{7}\right)^1$$

Alternative interpretation:

$$P(\text{one roll, one six}) = \frac{2}{3}$$

$$P(\text{at least 9 sixes}) = {}^{10}C_{10}\left(\frac{2}{3}\right)^{10} + {}^{10}C_9\left(\frac{2}{3}\right)^9\left(\frac{1}{3}\right)^1$$



parabola:  $y = bx - ax^2$

subst point  $(h, 20)$

$$20 = bh - ah^2$$

$$\therefore b = \frac{20 + ah^2}{h} \quad \text{--- ①}$$

subst point  $(h+3, 20)$

$$20 = b(h+3) - a(h+3)^2$$

$$20 = bh + 3b - ah^2 - 6ah - 9a \quad \text{--- ②}$$

subst eqn ① into eqn ②

$$20 = h\left(\frac{20 + ah^2}{h}\right) + 3\left(\frac{20 + ah^2}{h}\right) - ah^2 - 6ah - 9a$$

$$20 = 20 + ah^2 + \frac{60}{h} + 3ah - ah^2 - 6ah - 9a$$

$$0 = \frac{60}{h} - 3ah - 9a$$

$$3ah + 9a = \frac{60}{h}$$

$$a(3h + 9) = \frac{60}{h}$$

⑦ b) i) cont'd...

$$a = \frac{60}{h(3h+9)}$$

$$a = \frac{60}{3h(h+3)}$$

$$a = \frac{20}{h(h+3)}$$

subst into eqn ①

$$b = \frac{20 + \left(\frac{20}{h(h+3)}\right)h^2}{h}$$

$$hb = 20 + \frac{20h}{h+3}$$

$$hb = \frac{20(h+3) + 20h}{h+3}$$

$$b = \frac{20h + 60 + 20h}{h(h+3)}$$

$$b = \frac{40h + 60}{h(h+3)}$$

$$= \frac{20(2h+3)}{h(h+3)}$$

⑦ b) ii)  $x = Vt \cos \theta$

$$\therefore t = \frac{x}{V \cos \theta}$$

subst into  $y = Vt \sin \theta - 5t^2$

$$y = V \cdot \frac{x}{V \cos \theta} \cdot \sin \theta - 5 \left( \frac{x}{V \cos \theta} \right)^2$$

$$y = x \tan \theta - \frac{5x^2}{V^2 \cos^2 \theta}$$

b) iii) Equating coefficients of  $x^2$  in parts (i) & (ii)

$$\frac{20}{h(h+3)} = \frac{5}{V^2 \cos^2 \theta}$$

$$20V^2 \cos^2 \theta = 5h(h+3)$$

$$V^2 \cos^2 \theta = \frac{h(h+3)}{4}$$

iv) If  $\theta = 45^\circ$

$$\text{then } V^2 \cos^2 45^\circ = \frac{h(h+3)}{4}$$

$$V^2 \left( \frac{1}{2} \right) = \frac{h(h+3)}{4}$$

$$V^2 = \frac{h(h+3)}{2}$$

————— (A)

⑦ b) iv) cont'd...

Equating coeffs of  $x$  in parts (i) & (ii)

$$\tan \theta = \frac{20(2h+3)}{h(h+3)}$$

But  $\theta = 45^\circ$

$$\tan 45^\circ = \frac{20(2h+3)}{h(h+3)}$$

$$1 = \frac{20(2h+3)}{h(h+3)}$$

$$h^2 + 3h = 40h + 60$$

$$h^2 - 37h - 60 = 0$$

$$h = \frac{37 \pm \sqrt{37^2 - 4 \times 1 \times -60}}{2}$$

$$= \frac{37 \pm \sqrt{1609}}{2}$$

But  $h$  is a length and must be positive.

$$\therefore h = \frac{37 + \sqrt{1609}}{2}$$

$$= 38.556 \dots$$

Subst this into eqn (A)

$$v^2 = \frac{38.556 \dots (38.556 \dots + 3)}{2}$$

$$v^2 = 801.123 \dots$$

$$\therefore v = \pm \sqrt{801.123 \dots}$$

$$\therefore v = 28.304 \dots \quad (\text{as velocity} > 0 \text{ initially})$$